

Part VIII: Dynamics and Statistics of GA Relics – Foundations for Quantitative Predictions

Abstract: This part formalizes the hypothesis that dark matter consists of a cosmic population of sub-network relics locked in a White Hole (GA) regime. We define the parameters of a "hidden palace", derive its mass function and dynamic distribution, and explore the distinctive observational signatures of this granular dark matter model.

VIII.1 Quantitative Definition of a GA Relic ("Hidden Palace")

A GA relic is an autonomous sub-network, $\mathcal{R}$, characterized by:

1. Number of nodes: $N_{\mathcal{R}} \gg 1$.
2. Average internal coupling: $\bar{J}_{\text{int}} \approx J_c$.
3. Effective external coupling: $\kappa_{\text{ext}} \equiv \epsilon \bar{J}_{\text{int}}$, with $\epsilon \sim \exp(-S_{\text{horizon}}) \ll 1$.
4. Binding energy (emergent mass): $M_{\mathcal{R}} = N_{\mathcal{R}} m_0 - \frac{\bar{J}_{\text{int}}}{2} \sum \langle kl \rangle \in \mathcal{R} \langle \vec{\sigma}_k \cdot \vec{\sigma}_l \rangle$.
5. Emergent Schwarzschild radius: . The stability condition requires $R_{\mathcal{R}} > R_{s,\mathcal{R}}$.

VIII.2 Ensemble Statistics: The Mass Function and Cosmic Evolution

The initial scale-invariant distribution is modified by coalescence (relic merging) and fragmentation:

$$\frac{dn}{dM} \Big|_{\text{init}} = A_0 M^{-\beta}, \quad \beta \approx 2.$$

The processes of coalescence ($M_1 + M_2 \rightarrow M_{\text{new}}$) and fragmentation lead to a mature equilibrium distribution in galactic halos:

$$\frac{dn_{\text{halo}}}{dM} = \mathcal{N}_0 \left(\frac{M}{M_*}\right)^{-\alpha} \exp \left[-\left(\frac{M}{M_{\text{max}}}\right)^\zeta\right]$$

where $\alpha \in [1.8, 2.1]$, M_* is a characteristic mass, and M_{max} is the maximum stability mass.

VIII.3 Collective Dynamics and Derivation of the Density Profile $\rho_{VG}(r)$

We consider a spherically symmetric halo of relics in hydrostatic equilibrium. The dynamics are governed by:

$$\frac{d}{dr} [\rho(r) \sigma_v^2(r)] = -\rho(r) \frac{GM_{\text{tot}}(< r)}{r^2}$$

With a barotropic equation of state $P(r) = \kappa \rho(r)^\Gamma$ and $\Gamma \approx 1$, and density defined as an integral over the local mass function, the approximate analytical solution is a generalized cored profile:

- $\gamma \approx 0 - 1$ (soft/flat core exponent) – a result of the topological stability of relics.
- $\beta \approx 2 - 3$ (outer slope exponent).
- $\alpha \geq 1$ (transition parameter).

Comparison with Standard Profiles:

Profile	Shape (simplified)	Core (γ)	VG Compatibility
NFW (Λ CDM)		1 (cusp)	NO
Einasto		0 (flat)	PARTIAL
VG (Proposed)		0 - 1 (flat/soft)	YES

VIII.4 Dynamic Observational Signatures and Quantitative Tests

1. Galactic Rotation Curves:

· VG Prediction: Due to the flat core ($\gamma \approx 0$), the orbital velocity $V(r)$ increases linearly ($V \propto r$) in the central region, providing a better fit to observations in dwarf galaxies than the cuspy ($\gamma \approx 1$) of NFW profile from Λ CDM

· Calculation: , with $M(< r) = 4\pi \int \rho_{VG}(s)s^2ds$.

2. Gravitational Microlensing Profile:

· VG Prediction: The mass function with $a \approx 2$ leads to a spectrum rich in short-duration events. The distribution of event timescales t_E is:

$$\frac{d\Gamma}{dt_E} \propto t_E^{3-2\alpha} \approx t_E^{-1} \text{ (for } \alpha = 2\text{)}$$

This excess of short events is a falsifiable signature of VG granularity, testable with probes like LSST or Roman.

VIII.5 Conclusion and Testability

Part VIII transforms the "Hidden Palaces" hypothesis into a quantitative framework. The $\rho_{VG}(r)$ profile and associated mass function make three key predictions: (1) flat galactic cores, (2) characteristic rotation curve shapes, (3) a specific spectrum of short-timescale microlensing events. These provide clear pathways for testing and falsifying the model against precision cosmological observations.